

- 1 (a) Fig. 1.1 is a photomicrograph of a region of eukaryotic tissue. Some of the cells are in stages of mitosis.

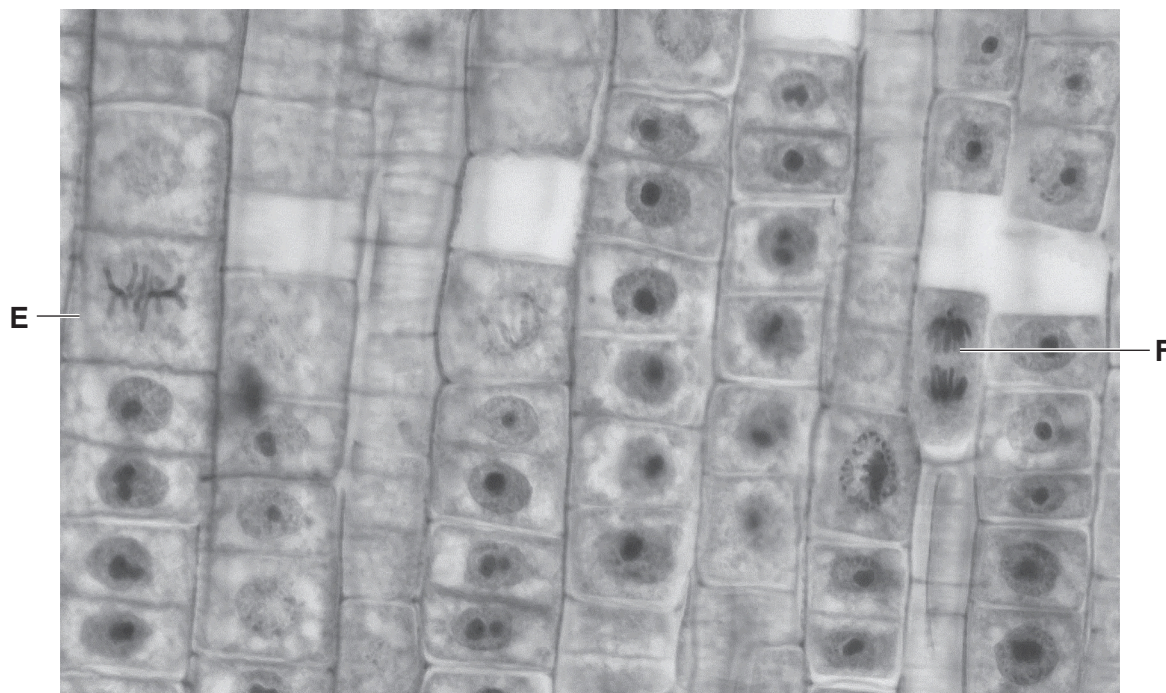


Fig. 1.1

- (i) Identify which stage of mitosis is shown in cell **E** and in cell **F** in Fig. 1.1.

E

F

[2]

- (ii) Microtubules are present within the cells that are in stages of mitosis, but these are not visible in Fig. 1.1.

State the function of microtubules in mitosis.

.....
.....
..... [1]

- (iii) State, with a reason, whether Fig. 1.1 shows a region of animal or plant tissue.

.....
.....
..... [1]

(b) Semi-conservative replication of DNA occurs during interphase, before mitosis begins.

Write the correct term in the spaces provided to complete each of statements **A** to **D**.

- A** The DNA double helix unwinds and is separated into two template strands when bonds holding the two strands together are broken.
- B** One of the template strands of DNA is copied in fragments. The enzyme is required to join the fragments together to form a continuous strand of DNA.
- C** Complementary DNA nucleotides are added to the template strands, catalysed by the enzyme
- D** are regions of repeating nucleotide sequences at the ends of chromosomes that allow the continued replication of DNA, without the loss of genes.

[4]

[Total: 8]